
INVESTMENT BRIEFING REPORT

South Australian Electricity Market

Capture Price Dynamics, Technology Economics, and Portfolio Recommendation



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*Prepared for an Energy Finance Client Considering Acquisition of a
Renewable Energy Developer Operating in the National Electricity Market*

ANALYSIS PERIOD:

1 July 2022 – 30 June 2024.

PREPARED BY:

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Executive Summary

The Investment Thesis

South Australia has become a structurally bifurcated electricity market where dispatch flexibility and timing earn premium revenue while raw generation volume does not.

- **Storage and modern flexible gas occupy the efficient frontier;**
- **Standalone solar is structurally compromised;**
- **Wind sits between them with the advantage of a deployable scale.**

The optimal portfolio for new capital deployment is a **wind-plus-storage hybrid weighted approximately 60/40**, supplemented by selective solar exposure only where firm contracts or co-located storage protect revenue.

Key Findings

Capture price spread: Capture prices in SA range from approx. **\$25/MWh (solar) to \$430/MWh (diesel)**, around a 17× spread that makes technology selection the dominant revenue driver.

Dispatchable premium widening: Capture factor for dispatchable assets has roughly doubled over two years, gas from ~1.7× to ~4.9× the time-weighted price; battery storage from ~2.7× to peaks of ~5.3×.

Solar cannibalization confirmed: Solar capture factor on a measurable downward trend (−0.026/year), reflecting structural alignment of solar dispatch with depressed midday prices.

Wind resilience: Wind capture factor rising (+0.106/year), with extended periods exceeding 1.0× the market average through late 2023 / early 2024.

Storage risk-adjusted dominance: Battery net revenue per MW competitive with the highest-earning gas peakers, at materially lower volatility (CV ~0.55–0.60 vs ~1.0–1.2 for gas).

Time-limited storage opportunity: Battery premium starting to compress at the right edge of the dataset. The early-mover advantage in storage is real but finite.

1. Market Context

1.1 SA Wholesale Spot Price Trajectory

The two-year period contains *one extreme event* and *one structural trend*. The July 2022 average of spot price approximately \$375/MWh, almost $4\times$ *the two-year mean* of \$101/MWh, reflecting the global gas price shock following the Russia-Ukraine conflict and this is not considerably a base-case condition for forward modelling. From November 2022 onward, prices *stabilized in a structurally lower band*, with the median post-shock month sitting closer to \$60–80/MWh.

The July 2023 price sinks to below \$100/MWh, which is more analytically significant for an investor. Demand during this window was comparatively high (~1,150 GWh per month), so *the decay was not demand-driven*. [AEMO's Q2 2023 Quarterly Energy Dynamics](#) reports attributes elevated SA prices in the period of May-Sep 2023 to wind underperformance and increased reliance on directed gas-fired generation for system security. South Australian system security direction costs rose from \$7.4 million in Q2 2022 to \$21.7 million in Q2 2023, with the average compensation price climbing from \$192/MWh to \$337/MWh.

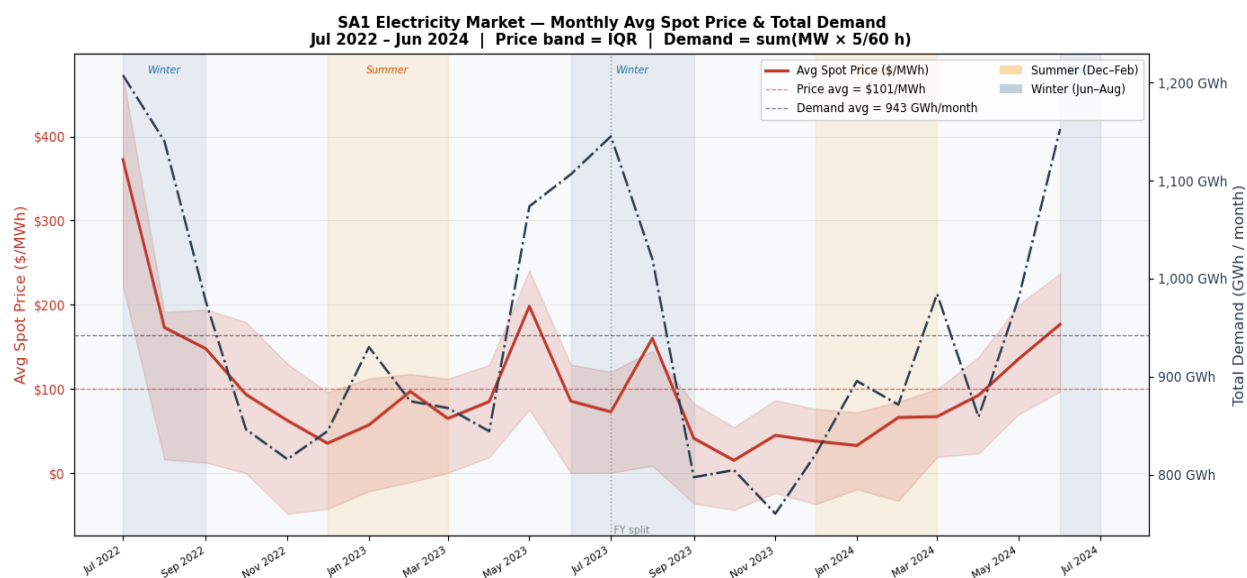


Figure 1.1 SA spot price normalized quickly after the July 2022 gas-shock spike

Monthly average spot price and total demand, SA1 region, Jul 2022 – Jun 2024. Price band shows an interquartile range of within-month spot prices.

Investor takeaway: When renewable resources (wind, solar, etc.) fail, dispatchable assets capture concentrated revenue. A renewable-only portfolio is structurally short of scarcity premium; a renewable-plus-storage portfolio captures it.

1.2 Generation Mix and Renewable Share

Total generation in South Australia by surveyed renewable resources declined from a July 2022 of approximately 150 GWh to a corresponding July 2024 peak of approximately 117 GWh, recording at a 22% reduction. This reflects continued rooftop PV displacement of grid-supplied demand, consistent with AEMO reporting that distributed PV growth has been outpacing underlying demand growth.

For the investment thesis, this means the SA market is not growing in volume, but the opportunity is in displacing existing thermal generation, not serving new loads.

*The two-year average renewable share was 56.6% (F.1.2), with a sink of approximately 33% in winter months and a peak of approximately 83% in October 2023. The renewable share reveals an insignificant role over this period. Inter floors have remained around 33–37% across all three winters in the dataset, indicating that *capacity additions have roughly matched demand without raising the structural floor.**

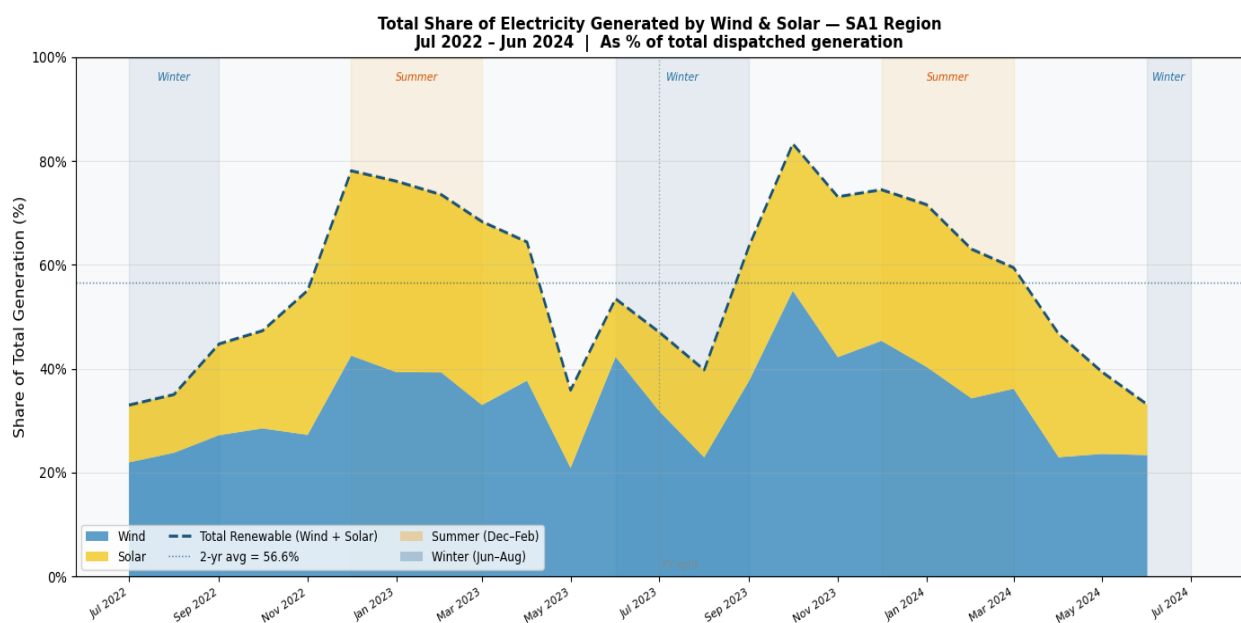


Figure 1.2 Total monthly generation declined 22% peak-to-peak as rooftop PV displaced grid demand

Monthly generation by technology with renewable share panel, SA1 region, Jul 2022 – Jun 2024.

2. Technology-Level Economics

2.1 Capture Price Distribution by Technology

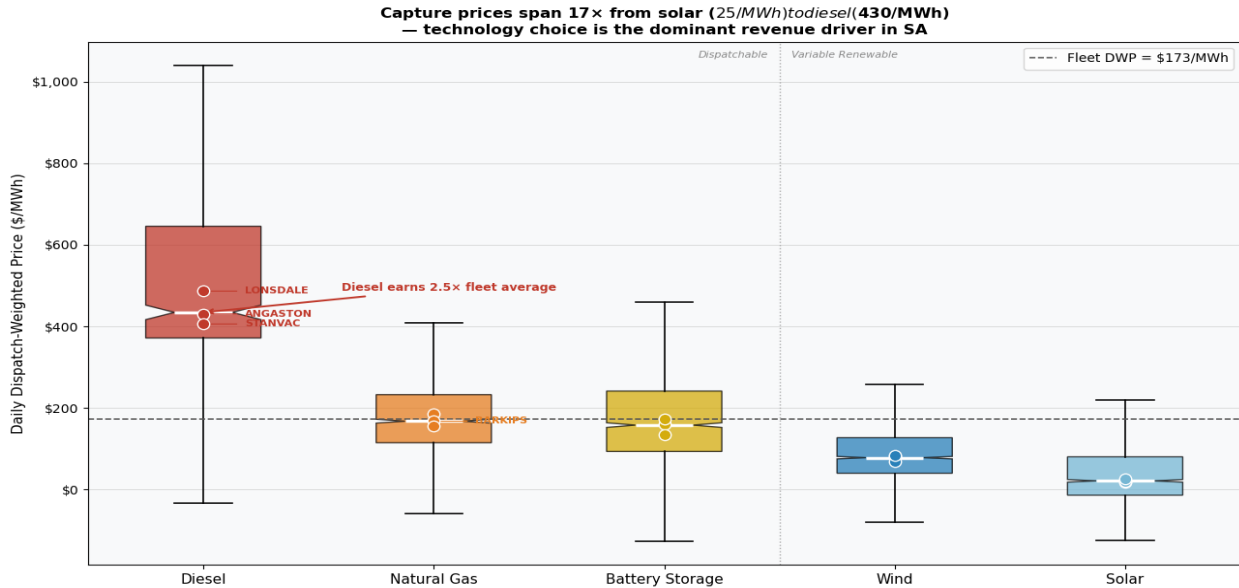


Figure 2.1 Capture prices across SA technologies shows that technology choice is the dominant revenue driver

Daily DISPATCH-WEIGHTED PRICE distribution by technology with per-DUID medians, Jul 2022 – Jun 2024.
Reference line at fleet DWP = \$173/MWh.

Three observations carry investment weight. **Diesel** earns *extreme median DISPATCH-WEIGHTED PRICE* through rare dispatch into market-cap intervals (F.2.1), but capacity is small, the operating profile is non-investable at scale. Storage and gas DWP are relatively equal at the technology median while batteries have already reached parity with gas peakers on a captured-price basis, despite battery storage being a relatively new asset class. **Solar's median DWP** of approximately \$25/MWh is roughly 15% of the fleet average and reflects Solar's near-perfect negative correlation with the SA daily price profile (T.2.1).

Technology	Median DWP (\$/MWh)	IQR (\$/MWh)	Multiple of Fleet DWP
Diesel	~430	370–650	2.5×
Battery Storage	~170	90–240	~1.0×
Natural Gas	~170	115–235	~1.0×
Wind	~70	40–125	0.4×
Solar	~25	0–80	0.15×

Table 2.1 — Capture price hierarchy by technology (fleet-weighted DWP across the period: \$173/MWh)

2.2 Within-Technology Dispersion

The largest within-technology dispersion is for the natural gas factory (F.2.2). BARKIPS records a daily DWP range of approximately \$16,700/MWh, nearly five times larger than QUARANTN (~\$3,600) and AGLHAL (~\$3,500). This reflects role disparity within the SA gas fleet: BARKIPS is a modern, fast-start reciprocating gas plant (commissioned 2019) capable of dispatching across both low and high price conditions, while the older open-cycle units operate in narrower peaking patterns.

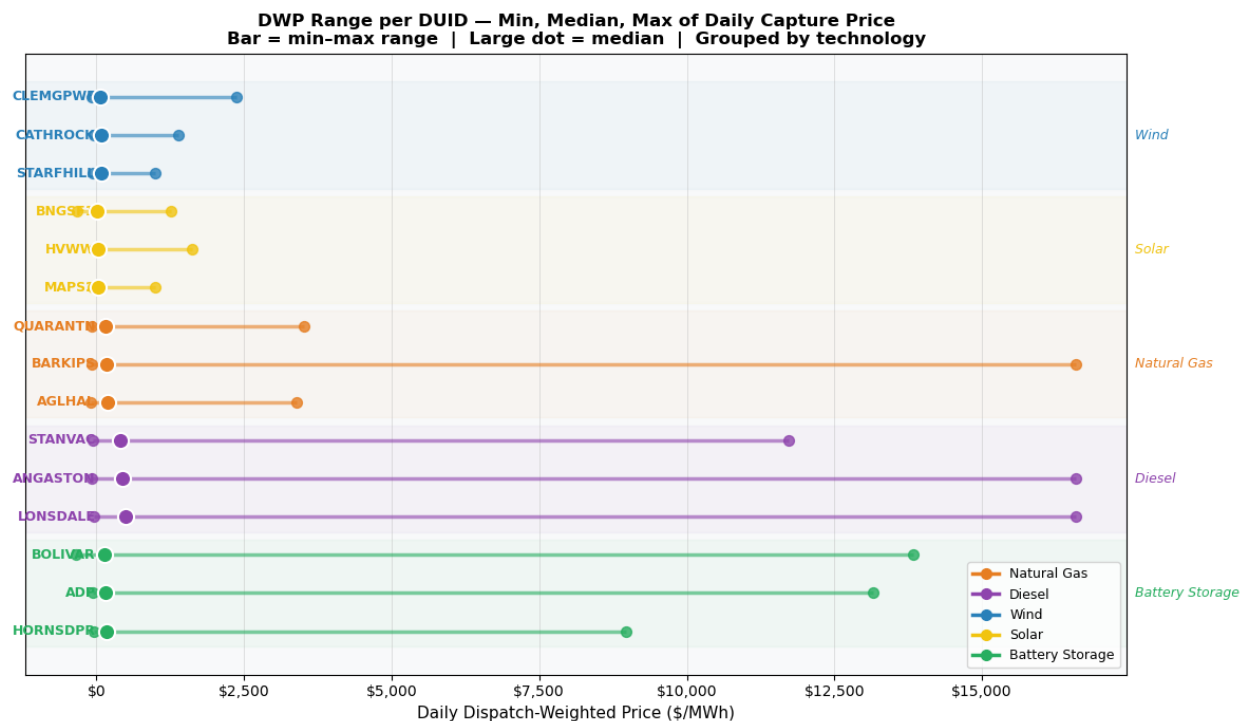


Figure 2.2 BARKIPS records the largest within-technology DWP range in the dataset

Daily DWP range (max – min) per generator, grouped by technology.

Investor takeaway: "Natural gas" is not a homogeneous asset class. DUID-level technical specifications materially affect revenue volatility. That's the reason why asset-by-asset due diligence is essential when calling an action.

2.3 Capture Factor Evolution

This is the single most economically significant chart in the analysis (F.2.3). The dispatchable panel shows **natural gas moving from ~1.7× in late 2022 to ~4.9× in early 2024**, and battery storage moving from ~2.7× to peaks of ~5.3×. Both then begin to decline by mid-2024, gas to ~1.7× and battery to ~1.6×. The variable renewables panel shows wind on a positive trend (+0.106/year), reaching periods above 1.0× in late 2023 / early 2024, while solar trends modestly negative (–0.026/year) and dips to near-zero in late 2023.

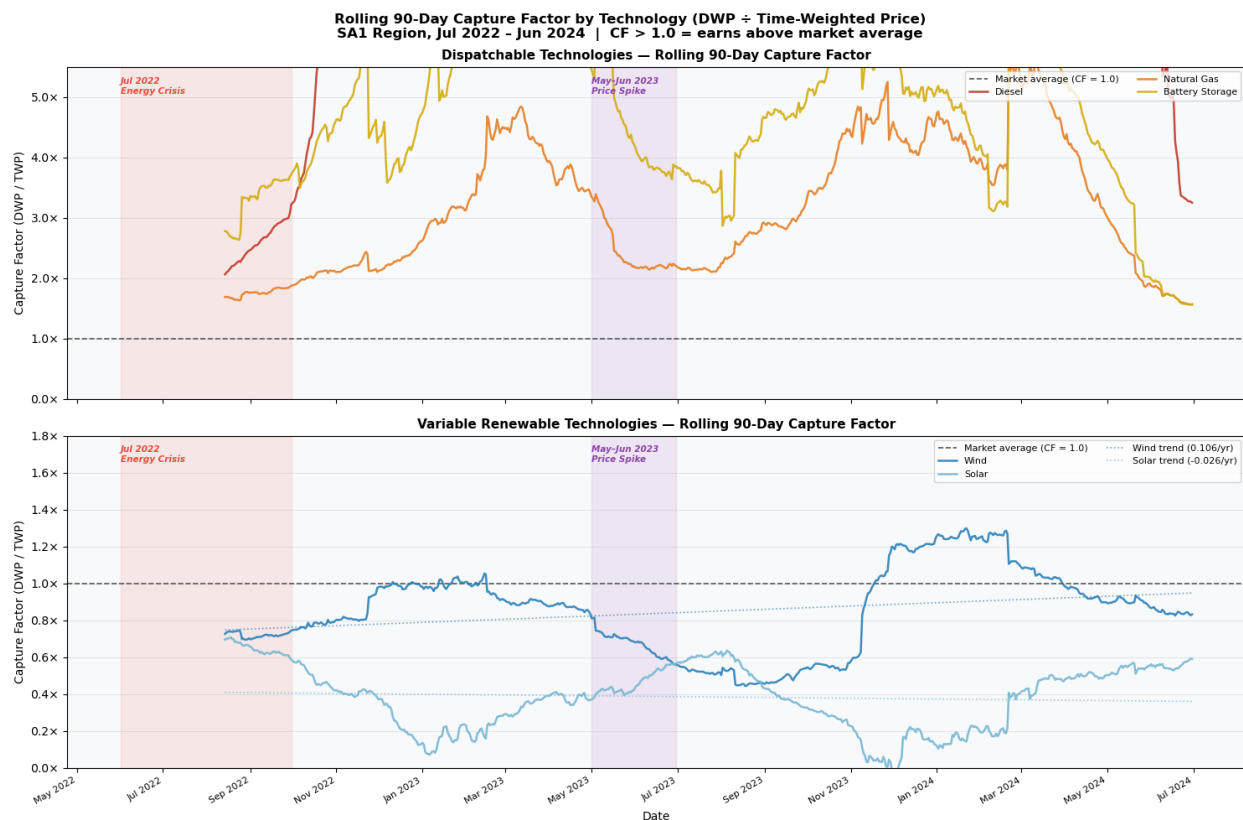


Figure 2.3 Dispatchable capture factor premium has roughly doubled over two years; storage premium starting to compress

Rolling 90-day capture factor (DWP ÷ TWP) by technology, SA1 region, dual-panel: dispatchable vs variable renewable.

Three Investment Implications

- **Dispatchable premium widening:** The premium that dispatchable assets earn is widening, not narrowing, as the system decarbonizes which directly contrary to the naive assumption that flexible thermal economics decline with renewable penetration.
- **Storage compression emerging:** The battery premium peaked in early 2024 and has begun to compress. This is the first empirical signal of storage saturation. Implying the window for outsized storage returns is real but finite.
- **Solar trend compounded over project life:** Solar capture compression of $-0.026/\text{year}$, compounded over a 15-year project life, implies a project capturing 60% of TWP today could capture only ~20% by 2040 if the trend continues.

2.4 Hourly Profile Mechanics

The mechanism behind capture price compression is visible in the hourly profile. SA spot prices average near \$0/MWh between 11:00 and 14:00 and peak at ~\$240/MWh at 18:00, a peak-to-trough ratio of approximately

240:1. Solar dispatch peaks at exactly the hour prices are lowest. Gas and battery dispatch peak in the evening when prices are highest.

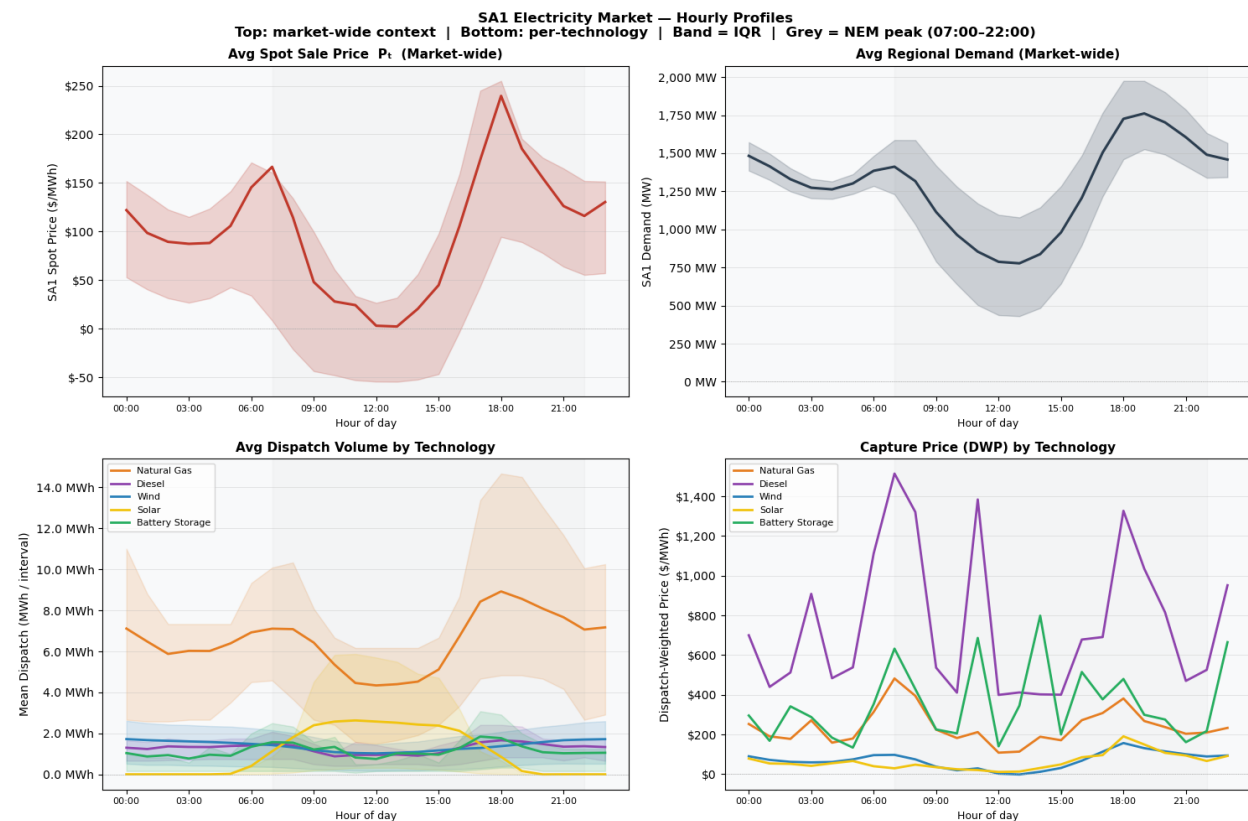


Figure 2.4 SA's daily price profile is bifurcated: midday near-zero, evening near \$240/MWh

Average hourly spot price, regional demand, dispatch volume by technology, and DWP by technology.

There is no operational fix for Solar's price-trough alignment within the technology itself, and critically it requires either co-located storage or firm contracting.

2.5 Capacity Factor (CF)

SA renewable assets are operating at benchmark, not underperforming:

Solar panels have a capacity factor of 10-20% of their maximum rated power per year. The carbon footprint of solar reaches 14–73-gram CO₂-e/kWh compared with 608 for gas and 975 for coal (Tawalbeh et al., 2020). Meanwhile, *wind farms performed better* in terms of capacity factor of 25-34% thanks to abundant wind resources, which meet the SA resource base support. However, the economic problem facing SA renewables is **not operational performance**; it is **actual price gaining**. In other words, the key solution for SA decarbonized systems is market design and capture-price dynamics, not operational execution. This is critical under the client's view, telling what kind of intervention should be conducted (storage co-location: combined solar farms and batteries, combined wind farms and batteries). The more recent wind farms have been combined with solar farms and batteries, such as the Port Augusta Renewable Energy Park (The South Australian Government Journal, n.d.).

Within-technology dispersion is large and signals the investment potential:

CLEM wind farm is out-performed corresponding to the other 2 counterparts. Within solar, **BINGSF1** is **nearly doubled HVWW**. These differences likely reflect site quality, equipment aging, inverter efficiency, and local network constraints. Asset level due diligence matters more than category-level assumptions.

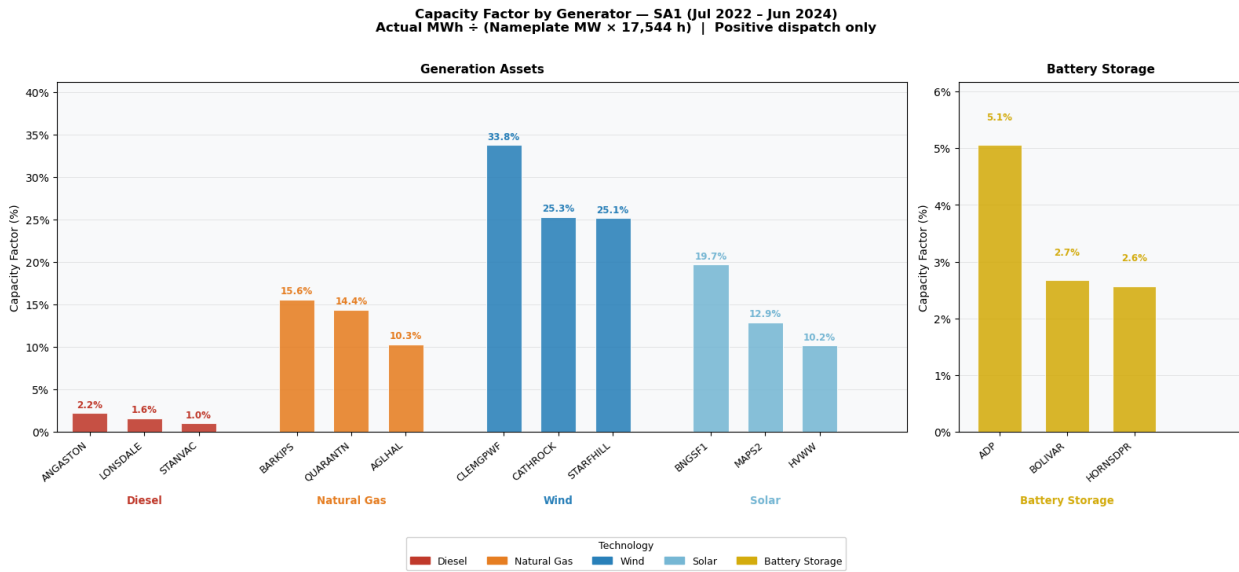


Figure 2.5 SA renewable capacity factors are at benchmark, not underperforming

Realized capacity factor by generator, Jul 2022 - Jun 2024. Battery capacity factor on a throughput basis (positive dispatch only).

Battery should be treated as a separately gaining stream:

ADP's 5.1% throughput CF translates to roughly 1.0–1.2 daily cycles for a typical 1–2-hour battery, which is high operational utilization by industry standards. The chart visually understates battery economic productivity because storage doesn't fit the energy-production paradigm. This is precisely why the risk-return scatter (Exhibit 3) treats batteries on a separate basis. In addition to residential battery systems, Australia installed 27 large-scale utility batteries in 2023, delivering a total storage capacity of 5 GW/11 GWh Clean Energy Council (CEC, 2024).

Technology	Realised CF Range	Industry Benchmark (SA)	Read
Wind	25–34%	28–38%	At benchmark
Solar (utility)	10–20%	22–28%	In-range to lower end
Natural Gas	10–16%	8–18%	Peaking/firming role
Diesel	1–2%	<3%	Emergency-only
Battery (throughput)	2.5–5.1%	N/A	Use cycles/day instead

Table 2.5 — Realized capacity factors against industry benchmarks

Critical framing: The economic problem facing SA renewables is **not operational underperformance**; assets are hitting their resource targets. The challenge is purely revenue, which means the relevant interventions are revenue contracting and storage co-location, not technology selection.

3. Risk-Adjusted Investment Recommendation

3.1 Revenue per MW

Wind revenue per MW oscillates between \$5,000 and \$25,000 per month. Solar between \$0 and \$15,000 per month. Battery (modelled arbitrage) between \$8,000 and \$42,000 per month. Despite battery's low throughput capacity factor, it converts price spreads into revenue more efficiently per MW than energy-driven renewables convert generation into revenue.

South Australia market is producing relatively similar revenue per MW across different generators. From late 2022 onward, most technologies revenue per MW all sit broadly in the 5k-25k/MW range for most months, except some occasionally spiking higher observed in gas. Outside of specific stress events, the revenue differentiation is minor. It's no longer about revenue on average, but about distribution shape. Gas earns its premium through occasional spikes; wind earns consistently and stably; solar earns very little either way. This is further explained in the risk-return scatter plot (Fig 3.2).

Scarcity premium accrues to dispatchable assets, not variable renewables. August 2023 saw the most significant signal for this assumption. When wind drought is visible in August, wind revenue holds at moderate level (\$15,000), Gas, diesel and battery revenue also jump into the highest momentum across the 2-year window, but solar barely benefits. This states that any portfolio without any dispatchable exposure does not participate in the scarcity premium that the SA currently delivers.

Renewables earn volume; batteries earn timing and in the SA market, timing is currently more valuable.

Monthly Revenue per MW — SA1 Generators (Jul 2022 - Jun 2024)

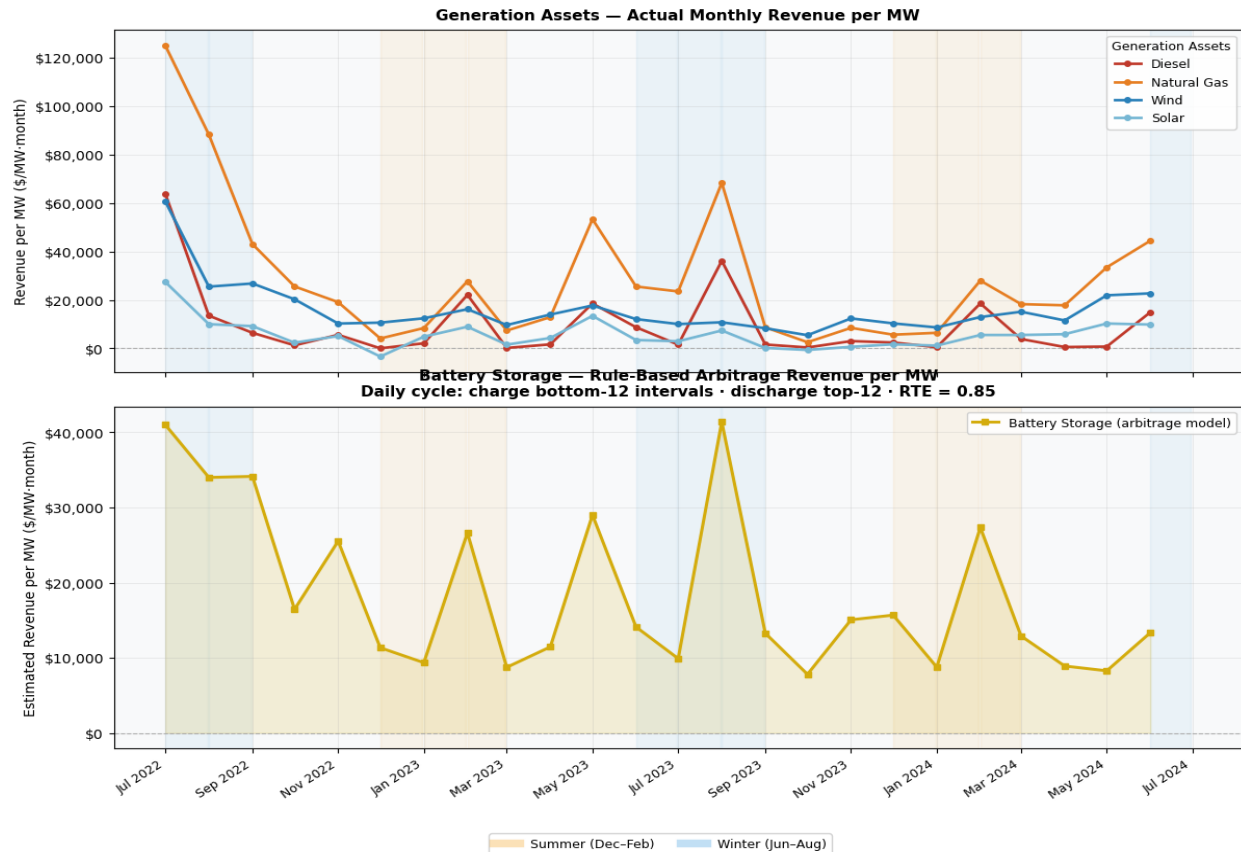


Figure 3.1 Battery net revenue per MW is competitive with wind and dramatically higher than solar

Monthly revenue per MW, SA1 generators (battery on rule-based arbitrage model: charge bottom-12 intervals, discharge top 12, RTE = 0.85).

3.2 Risk-Return Synthesis

Their mean revenue per MW is around 18,000 sits above the wind cluster and well above solar, while their CV is the lowest on the chart (0.59). This implies that storage is the single most safety technology in the SA market. However, the modest capacity factor (below 5%) reflects the efficient frontier composed of relatively small-scale assets. Deploying significant capital into storage requires new construction, not buying it.

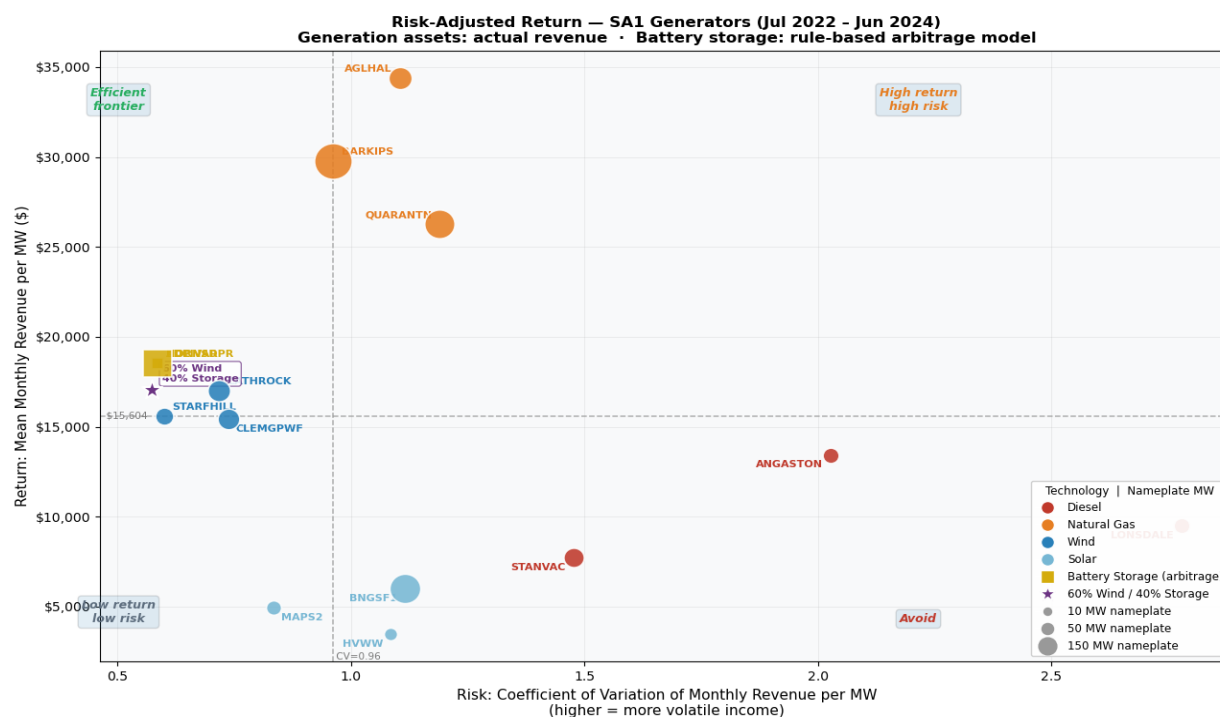


Figure 3.2 Storage occupies the efficient frontier; the wind-storage hybrid portfolio is the deployable representation

Mean monthly revenue per MW vs coefficient of variation, SA1 generators. Marker size = nameplate capacity. Quadrant lines at fleet medians. Hypothetical 60/40 wind storage portfolio overlaid as star marker.

Asset Class	Mean Revenue (\$/MW/month)	CV	Position
Battery cluster	~\$18,000	~0.55–0.60	Efficient frontier
Wind cluster	~\$15,500–17,000	~0.60–0.75	Middle ground
Gas peakers	~\$26,500–34,500	~1.0–1.2	High return / high risk
Diesel	~\$8,000–13,500	~1.5–2.7	Avoid
Solar	~\$4,000–6,000	~0.85–1.1	Low return / mod risk
60/40 Wind-Storage Portfolio	~\$17,000	~0.55	On efficient frontier with deployable scale

Table 3.2 — Risk-return profile by asset class with hybrid portfolio overlay

Gas standalone is at the highest risk, with the highest estimated revenue per MW. For a risk-tolerant investor, gas delivers higher absolute returns; for a risk-adjusted investor, storage is the better technology, potentially dealing with the scarcity of system decarbonization. As Chris Bowen, the Minister for Climate Change and

Energy said, “if we have further transmission delays, batteries become more and more useful for every year of delay, ...we can use batteries in the near term to decongest the existing network.” (*MacDonald-Smith, 2023*).

Wind is the most scalable acceptable investment in the SA market (meaningful nameplate capacity, delivering respectable returns. Solar, by contrast, exhibits the lowest revenue due to the surplus supply but moderately high volatility - the worst single portfolio.

The hypothetical 60% wind - 40% storage portfolio sits inside the frontier. The portfolio star market sits at approximately \$17,000/MW mean revenue with CV ~0.55 (slightly lower mean than pure storage but at marginally lower risk) and with the critical advantage of *deployable scale*. Wind farms have larger nameplates than batteries, so a 60/40 mix can absorb meaningful capital allocations while preserving the risk-return characteristics of the storage-dominant frontier. **This is the investable version of the efficient frontier:** storage alone is too small to absorb large capital, but a wind-storage hybrid retains most of the risk-adjusted attractiveness while becoming deployable at an institutional scale.

3.3 Recommendations

RECOMMENDATION 1: HIGHEST CONVICTION

Storage as core technology, deployed within 24–36 months

Battery storage occupies the efficient frontier with the lowest revenue volatility and competitive returns. The capture-factor premium has begun to compress, indicating a finite early-mover advantage. The client should priorities storage exposure aggressively in the near term while the premium remains intact.

RECOMMENDATION 2: SCALE PLAY

Wind for deployable scale, with asset-level diligence

CLEMGPWF’s capacity factor (33.8%) is materially higher than peers, illustrating that within-technology dispersion is large enough to justify careful asset selection. Wind is the renewable technology that can absorb meaningful capital without sacrificing risk-adjusted returns, but specific asset selection matters more than the category-level allocation decision.

RECOMMENDATION 3: AVOID

Avoid standalone merchant solar

The combination of low capture price, low revenue per MW, moderate volatility, and structural cannibalization makes merchant solar fundamentally not investable in SA based on this two-year window. Any solar exposure must come either with firm revenue contracts (PPA, CIS underwriting) or with co-located storage that shifts dispatch into evening high-price hours.

RECOMMENDATION 4: TACTICAL

Gas as tactical, exit-defined position

Modern flexible gas (BARKIPS-class) earns the highest absolute revenue per MW in the dataset but at materially higher risk and with long-term decarbonization exposure. Treat as tactical yield enhancement with a defined exit horizon tied to storage capacity build-out milestones, not a strategic core holding.

3.4 Risk Register

The recommendation above is supported by the two-year analytical window. Forward modelling should explicitly stress-test the following risks, any of which could materially alter the conclusions:

- **Storage saturation:** Faster-than-modelled compression of battery capture factors as the SA storage queue commissions through 2025–2027.
- **Policy reversal:** Federal Capacity Investment Scheme or state policy changes reducing contracted revenue floors for new renewables.
- **Connection delays:** Connection queue and transmission build delays affecting deployment timelines and IRR realization.
- **Gas price volatility:** Movements in domestic gas prices affect peaker economics and the broader price formation in low-VRE periods.

- **Resource scarcity events:** Multi-month wind or solar droughts producing concentrated revenue events that the two-year window may under-represent.

Methodology and Sources

Data Sources

SA1 spot price and dispatch data at 5-minute resolution for 15 generators (3 each of diesel, gas, wind, solar, battery storage) over the period 1 July 2022 - 30 June 2024. Nameplate capacities sourced from the AEMO Generation Information page. Industry context drawn from AEMO Quarterly Energy Dynamics reports (Q2 2023, Q3 2025) and the 2025 Electricity Statement of Opportunities.

Capture Price (Dispatch-Weighted Price)

Capture price computed using the standard formula: $DWP = \Sigma(P_t \times Q_t^s) \div \Sigma Q_t^s$, where P_t is the SA1 spot price at 5-minute interval t , and Q_t^s is the dispatch of station s at interval t . Capture factor defined as $DWP \div$ time-weighted average spot price over the relevant rolling window. Rolling 90-day windows used to smooth short-duration noise.

Battery Treatment

Battery revenue calculated as net (discharge revenue minus charging cost) using rule-based arbitrage modelling (charge bottom-12 daily intervals, discharge top 12, round-trip efficiency = 0.85). Battery capacity factor and DWP are presented separately from energy-driven generators where methodologically appropriate, reflecting the structural difference between arbitrage-driven and energy-production-driven revenue.

Limitations

- **Window length:** Two years is a short window for forward inference; July 2022 is an outlier event distorting average. Sensitivity tests are recommended for any forward modeling.
- **Excluded data:** Dataset does not include forward project pipelines, contracted revenue mechanisms (PPAs, CIS contracts), curtailment of causation data, or marginal loss factors.
- **Battery model assumption:** Rule-based arbitrage model is a first-order approximation; actual battery operations involve FCAS revenue stacking, system strength services, and contract structures not captured here.

End of Report